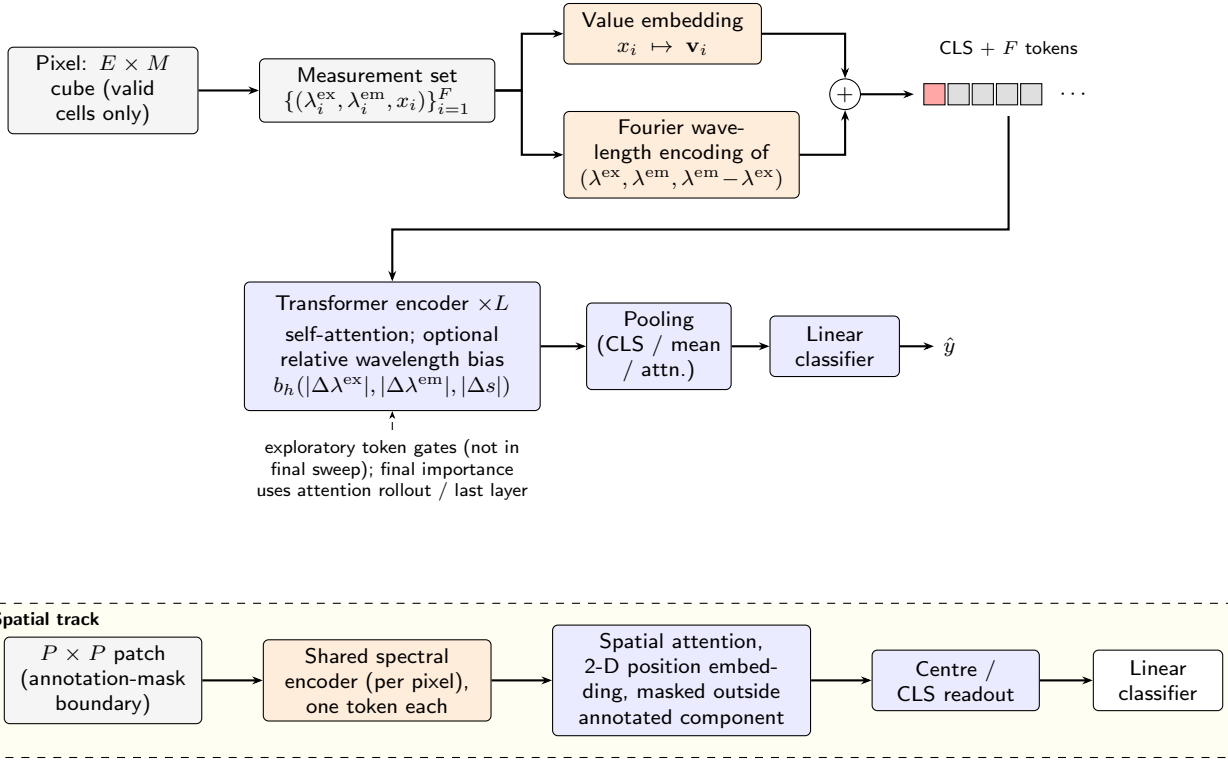

Pixel: $E \times M$ cube (valid cells only)
Measurement set $\{(\lambda_i^{\mathrm{ex}}, \lambda_i^{\mathrm{em}}, x_i)\}_{i=1}^{F}$
Value embedding $x_i \mapsto \mathbf{v}_i$
Fourier wavelength encoding of $(\lambda^{\mathrm{ex}}, \lambda^{\mathrm{em}}, \lambda^{\mathrm{em}} - \lambda^{\mathrm{ex}})$
CLS + F tokens
Transformer encoder $\times L$
self-attention; optional relative wavelength bias $b_h(|\Delta\lambda^{\mathrm{ex}}|, |\Delta\lambda^{\mathrm{em}}|, |\Delta s|)$
exploratory token gates (not in final sweep); final importance uses attention rollout / last layer
Pooling (CLS / mean / attn.)
Linear classifier
$\hat{y}$
Spatial track
$P \times P$ patch (annotation-mask boundary)
Shared spectral encoder (per pixel), one token each
Spatial attention, 2-D position embedding, masked outside annotated component
Centre / CLS readout
Linear classifier